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Strain in Pipe Elbows Due to Wave Propagation Hazards

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Abstract

Wave propagation hazards result from traveling waves that induce out-of-phase motion along a pipe axis. These in turn result in the generation of axial and flexural strains in buried pipe. For straight sections of buried pipe, axial strains are typically an order of magnitude larger than the wave propagation induced flexural strains. However at an elbow, the reverse occurs. By the nature of an elbow, the axial strains are comparatively small while the flexural strains are comparatively large.

This paper presents calculated axial and flexural strains at an elbow in a buried continuous pipe subject to wave propagation. For most cases the flexural strains at the elbow are less than the corresponding wave propagation induced ground strains. Only for piping systems containing relatively thick walled pipe exposed to moderate wavelengths does the elbow flexural strain exceed the ground strain. That is, for most piping systems containing an elbow, the ground strain serves as an upper bound for the generated strain at an elbow due to wave propagation hazards.

Introduction

Experiments performed by Katayama (1988) concerning pipe soil interaction for a instrumented piping system containing a elbow are depicted in Figure 1. Observations for wave propagated generated pipe strains revealed high axial pipe strains away from the elbow, low axial strains at the elbow, high flexural strains at the elbow, low flexural strains away from the elbow, and the maximum axial strains (~17 ustrain) larger then peak bending strains (~5 ustrain). Procedures exist for the prediction of axial generated strains due to wave propagation effects away from the elbow, but few exist that address the elbow itself.

Background

The analytical model discussed herein represents a piping system with a 90 degree elbow and a R-wave propagating parallel to one of the pipe legs. The pipe leg oriented parallel

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to the wave propagation direction will be referred to as the longitudinal leg, while the leg perpendicular to the wave propagation direction is the transverse leg. Such a system is depicted in Figure 2, where the horizontal (longitudinal) component of the R-wave is moving parallel to the longitudinal leg of the piping system. The situation considered by Shah and Chu (1974) corresponds to a constant level of ground strain distributed uniformly over the longitudinal leg of the piping system. The situation considered by Shinozuka and Koike (1979) corresponds to ground strain distributed spatially as a sine wave over the longitudinal leg of the piping system. Specifically, Shinozuka and Koike consider the point in time when a ground displacement node point exists at point A of Figure 2. That is, the ground displacement is a minimum at point A of Figure 2 while the ground displacement at point B (i.e., a quarter wavelength away) is a maximum value, $u_g(\lambda/4)$, acting in the direction of WP (a quasi-static representation). McLaughlin (2003) also considered that the ground strain was distributed spatially as a sine wave over the longitudinal leg of the piping system.

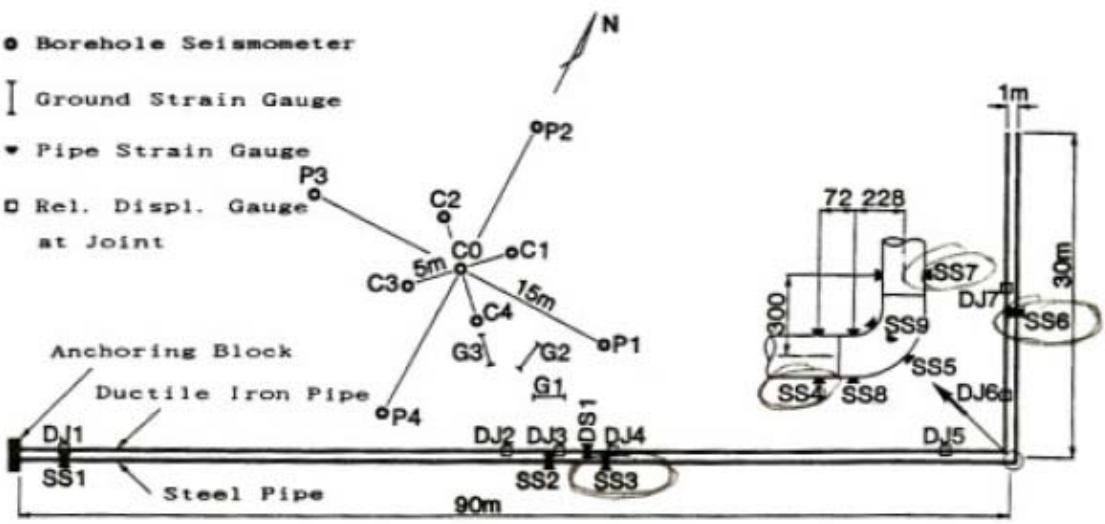

Fig. 1 General Layout of Observation System

For an infinitely stiff soil medium or for a very flexible pipe system, the pipe elbow as well as the transverse leg would follow the soil motion exactly and would have a displacement in the longitudinal direction of $u_g(\lambda/4)$. That is, the transverse leg would be straight and strain free. However, since the soil medium is not infinitely stiff, the elbow displacement will be somewhat less than $u_g(\lambda/4)$ whether or not there is soil slippage (i.e., slippage at the soil pipe interface) along the longitudinal leg. The difference between the actual elbow displacement and $u_g(\lambda/4)$ (=Δ) is designated Δ' in Figure 2. At the same time, the section of the transverse leg that is unaffected by the presence of the elbow will have a displacement in the direction of WP of $u_g(\lambda/4)$. The existence of such relative movement (i.e., Δ') is accommodated by the longitudinal extension of the longitudinal leg along with flexure of the transverse leg, or as the following.

$$\Delta' = \Delta - \Delta_e \tag{1}$$

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For the Shah and Shinozuka analysis methodologies, the longitudinal (i.e., axial) soil resistance along the longitudinal leg for the system in Figure 2 is modeled as a spring-slider, the transverse soil resistance for both pipe legs is modeled by an elastic soil spring (K_T), and any soil resistance in the longitudinal direction along the transverse leg is ignored. Finally, the pipe itself is elastic. McLaughlin's methodology contains soil resistance along the longitudinal leg in the longitudinal direction as well as spring-slider soil resistance along the transverse leg in the transverse direction. Soil resistance in the remaining (2) directions remain elastic, as well as the pipe itself, as shown in Figure 2.

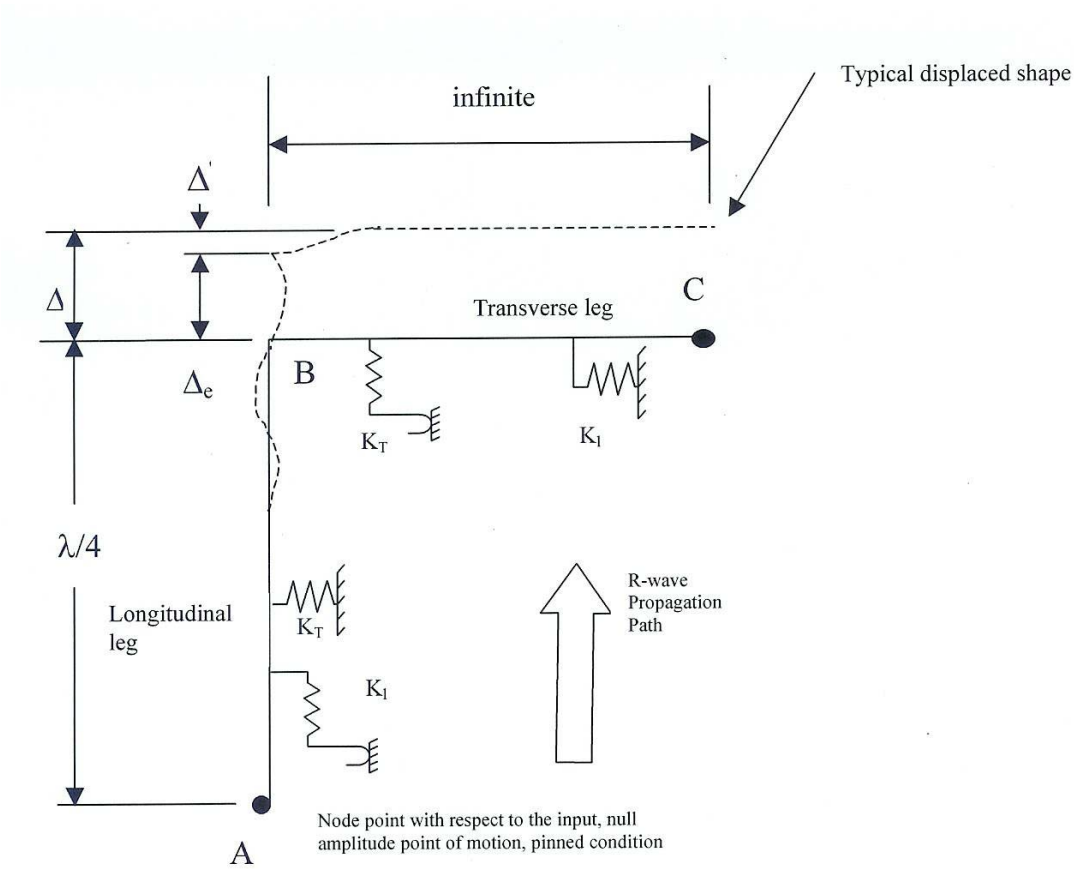

Figure 2. Proposed Model with Typical Displaced Shape of Pipe Elbow System Due to R-Wave Excitation

Once calculated, the relative movement at the elbow (the elbow transverse displacement), Δ', can be used to determine the internal forces at the elbow (axial force in the longitudinal leg (S), shear in the transverse leg (S2), and elbow bending moment (M)). These forces are related to the elbow transverse displacement, through load and displacement compatibility. For a beam on a elastic foundation, these relationships are defined as the following.

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$$\Delta' = \frac{4\lambda'}{3K_T} S \tag{2}$$

$$M = \frac{S}{3\lambda'} \tag{3}$$

$$S_2 = \frac{S}{3} \tag{4}$$

Where 1/λ' represents a characteristic length and is a function of the elastic transverse soil resistance (K_T) and the flexural rigidity of the beam, EI, as follows

$$\lambda' = \sqrt[4]{K_T / 4EI}$$

These relationships will be redefined for a beam on a inelastic foundation by McLaughlin in the next section. What is unique to the methodologies discussed herein is the manner in which Δ' is defined (essentially, how the resistance of the elbow is represented).

Shah and Chu

Shah and Chu define Δ' as the following,

$$\Delta' = \varepsilon_g L' - \frac{SL'}{AE} - \frac{t_u L'^2}{2AE} \tag{5}$$

where S is the axial force in the elbow. The first term on the right hand side represents the amount of soil movement at the elbow due to a uniform ground strain acting over an effective slippage length, L'. The second term represents the longitudinal displacement at the elbow due to the axial load, S, also acting over a distance L'. The third term is the elbow displacement due to soil frictional forces, t_u , in the direction of WP also acting over a distance L'. The maximum axial force in the pipe is due to the axial force, S, at the elbow and the constant soil frictional force, t_u, over the length L', or

$$F_{\max} = S + t_u L' \tag{6}$$

Substituting Equation (2) and Equation (6) into Equation (5) yields

$$\frac{t_u L'^2}{2AE} + \left(\frac{4t_u \lambda'}{3K_T} - \frac{F_{\max}}{AE} + \varepsilon_g \right) L' - \frac{4F_{\max}\lambda'}{3K_T} = 0 \tag{7}$$

From the pinned boundary condition L' away from the elbow assumed in the model of Shah and Chu, the axial strain in the pipe at L' from the elbow matches the ground strain, hence

$$F_{\max} = \varepsilon_g AE \tag{8}$$

Substituting this expression into Equation (7) and simplifying yields an expression for L'

$$L' = \frac{4}{3}\Omega \left(\sqrt{1 + \frac{3F_{\max}}{2t_u \Omega}} - 1 \right)$$

where

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$$\Omega = \frac{AE\lambda'}{K_T}$$

With L' known, Equation (6) is used to solve for the axial force, S, in the elbow, while Equation (3) can then be used to find the corresponding moment, M in the elbow.

Shinozuka and Koike

Shinozuka and Koike assumed that the mechanism of slip between the pipe and the soil is assumed to involve the shear stress acting on the pipe surface rather than the force proportional to the relative displacement between the pipe and the soil, $K_l(u_g\text{-}u_p)$. The displacement at the elbow in the direction of WP is then defined as

$$\Delta' = \varepsilon_g \frac{\lambda}{2\pi} - \beta\varepsilon_g \frac{\lambda}{2\pi} - \frac{\lambda S}{4AE} \tag{9}$$

where β is defined as a conversion factor that accounts for the degree of longitudinal slip that has occurred. The first term is the net soil displacement observed at the pipe elbow when accounting for longitudinal slip, while the second term represents the elongation of the longitudinal leg due to the axial force, S. The variable β is defined as follows.

$$\beta = q\frac{\gamma_{cr}}{\gamma_o}\frac{1}{1+\left(\frac{2\pi}{\lambda}\right)^2\frac{AE}{K_l}} \tag{10}$$

where q is a factor which ranges from 1 to π/2 and quantifies the degree of slippage at the pipe-soil interface, γ_{cr} represents the critical shear strain of soil at which slippage initiates (taken to be .001 strain), γ_o represents the maximum shear strain in the soil at the pipe soil interface while γ_o is defined as

$$\gamma_o = \frac{2\pi Et}{\lambda G}\varepsilon_g\beta_o$$

where G is the shear modulus of the soil and t is the pipe wall thickness. In Equation (10), the equivalent soil spring constant per unit area, K_l, is defined as

$$K_l = G$$

and β_o represents a conversion factor between the ground and the pipe strain and can be expressed as

$$\beta_o = \frac{1}{1+\left(\frac{2\pi}{\lambda}\right)^2\frac{AE}{K_l}}$$

Knowing the value of the conversion factor, β, Equations (2) and (9) are used to solve for the axial force, S, in the elbow. Equation (3) is then used to find the corresponding moment, M, in the elbow.

McLaughlin

The governing equations are developed for each of the pipe legs in Figure 2 independently. The seismic excitation is modeled by a sine wave spatial displacement distribution of the soil in the longitudinal direction. The resistance from the soil in the

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longitudinal direction is modeled as an elastic perfectly plastic spring (load-deformation relation of the soil is linear up to the shear resistance limit, beyond that point, the soil resistance is constant as displacements increase), while the pipe is modeled as an elastic element. Hence, given a sufficiently large ground strain, there will be two distinct regions along the longitudinal leg. For Region I, the soil spring/slider behavior is elastic, and the relative displacement between the soil and pipe is less than x_u. In Region II, the relative displacement between the pipe and soil exceeds x_u, and the soil resistance is a constant force per unit length, t_u. The length of the first region is designated x_{slip} and hence the length of the second region is $\lambda/4 - x_{slip}$. Hence, for full slip over the entire length of the longitudinal leg, $x_{slip} = 0$, while for no slip, $x_{slip} = \lambda/4$. By representing each region of pipe as a 2nd order boundary value problem, with compatibility imposed at the boundary (i.e., continuity of pipe displacement and pipe strain), the resulting expression for x_{slip} is generated, or

$$x_{slip} = \sin^{-1}\left(\frac{x_u + \frac{P}{\beta AE} - \frac{A_o \varepsilon_g}{\beta}\cos\left(\frac{2\pi x_{slip}}{\lambda}\right)}{(1 - A_0)\varepsilon_g\left(\frac{\lambda}{2\pi}\right)}\right)\left(\frac{\lambda}{2\pi}\right) \tag{11}$$

where, the displacement due to frictional forces and axial load, Δ_e, are defined by the following expression,

$$\begin{aligned}\Delta_e &= [S/AE + t_u(\lambda/4)/2AE]\lambda/4 + \varepsilon_g(\lambda/2\pi)A_O \sin(2\pi x_{slip}/\lambda) \\ &+ (P/\beta AE)\sinh\beta x_{slip}/\cosh\beta x_{slip} + \frac{t_u}{2AE}x_{slip}^2 - \frac{S}{AE}x_{slip} - \frac{t_u\lambda}{4AE}x_{slip} \\ &- \frac{A_O\varepsilon_g\cos\left(\frac{2\pi x_{slip}}{\lambda}\right)}{\beta\cosh(\beta x_{slip})}\sinh(\beta x_{slip})\end{aligned} \tag{12}$$

For the transverse leg, Region I represents the region in which transverse soil slippage has occurred and the pipe behaves as a Beam on an Inelastic Foundation (BIEF) over a distance of L_p. Region II represents the region in which no transverse soil slippage between the soil and the pipe has occurred and hence the pipe behaves as a Beam on an Elastic Foundation (BEF). Each region of the transverse section of pipe is represented as a 4th order boundary value problem, with compatibility imposed at the boundary (i.e., continuity of pipe displacement and pipe strain), the resulting expression for L_p is generated, or

$$L_p = \frac{-\left(2\lambda'^2 S - 2\lambda' P_u\right) \pm \sqrt{\left(2\lambda'^2 S - 2\lambda' P_u\right)^2 - 4\left(-\lambda'^2 P_u\right)\left(2\lambda' S + 2\lambda'^2 M - K_T y_u\right)}}{-2\lambda'^2 P_u} \tag{13}$$

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while, the displacement at the left edge of the BIEF, Δ', can be expressed as the following,

$$y(0) = \Delta' = 2V_1\lambda' / K_T + 2M_1\lambda'^2 / K_T + 2V_1\lambda'^2 L_p / K_T + 4M_1\lambda'^3 L_p / K_T - P_u L_p{}^4 / 8EI$$
$$+ SL_p{}^3 / 3EI + ML_p{}^2 / 2EI \qquad (14)$$

where,

$V_1 = S - P_u L_p$ and $M_1 = -P_u L_p^2/2 + SL_p + M$

The relationships for the axial displacement of the longitudinal leg (Equation (12), with unknowns S and x_{slip}), along with the relationship for the end displacement and end rotation for the transverse leg (Equation (14), with unknowns S, M and L_p) can be used to generate the expressions for a elbow on a inelastic foundation using load and displacement capatibility at the elbow. The resulting expression for the bending moment at the elbow is then,

$$M = \frac{-bS - c}{a} \qquad (14)$$

With a, b and c defined as the following.

$$a = -8\lambda'^3 / K_T - L_p / EI + 4\lambda'^4 / [K_T^2 \left(2\lambda' / K_T + 1/\beta AE\right)]$$

$$b = -2\lambda'^2 / K_T - 4\lambda'^3 L_p / K_T - L_p / 2EI$$

$$c = 2\lambda'^2 P_u L_p / K_T + 4\lambda'^3 P_u L_p{}^2 / 2K_T + P_u L_p / 6EI$$

Relation for the axial force, S, at the pipe elbow explicitly in terms of the pipe elbow transverse displacement, Δ', yields

$$S = \frac{\Delta' + \frac{ec}{a} - f}{d - \frac{eb}{a}} \qquad (15)$$

where d,e and f are as follows.

$$d = 2\lambda' / K_T + 2\lambda'^2 L_p / K_T + 2\lambda'^2 L_p / K_T + 4\lambda'^3 L_p{}^2 / K_T + L_p{}^3 / 3EI$$

$$e = 2\lambda'^2 / K_T + 4\lambda'^2 L_p / K_T + L_p / 2EI$$

$$f = -2\lambda' P_u L_p / K_T - 3\lambda'^2 P_u L_p{}^2 / K_T - 2\lambda'^3 L_p{}^3 P_u / K_T - P_u L_p{}^4 / 8EI$$

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Elbow Strains Due to R-Wave WP

A series of parametric studies were performed for a base configuration containing a 0.6m diameter steel pipe with a 0.012m wall thickness, buried about 0.9m in sandy soil with a unit weight of 1875 kg/m^3 and a soil shear angle of 35 degrees, corresponding to a pipe with a smooth, relatively hard coating. The maximum pipe elbow strains were generated by combining the contribution from the moment, M, along with the contribution from the force, S, at the elbow

$$\varepsilon_e = \frac{Mc}{EI} + \frac{S}{AE}$$

where I is the area moment of inertia of the pipe. The parameters evaluated include pipe diameter (.31m, .61m, .91m), pipe wall thickness (.004m, .006m, .012m), pipe burial depth (.91m, 1.2m, 1.5m) and WP wavelength (500m, 1500m, 3000m). The ground strain was varied up to a maximum value of 0.2%. For each case studied, only a single parameter was altered relative to the base case.

For the cases involving small wavelengths (500m), peak elbow strains approach the ground strain for increasing levels of ground strain, regardless of pipe wall thickness, pipe diameter or burial depth as depicted in Figure 3. For moderate wavelengths (1500m), similar trends exist, as shown in Figure 4, except for a relatively thick walled pipe (peak elbow strains ~ 1.25 ground strain). For large wavelengths (3000m), the level of ground strain necessary to generate slip and consequently large elbow strains is almost beyond the maximum level of ground strain examined (Figure 5). For all cases examined, the point at which slip is initiated occurs at the first inflection point of each elbow strain plot. As the wavelength is increased, the point is moved farther to the right, and consequently is not a limiting scenario. Furthermore, the location of the inflection point also depicts the maximum amount of axial strain that exists in the portion of straight pipe away from the elbow.

Summary

The presence of an elbow within a piping system acts to alter the response of such a system when subjected to earthquake effects. Since an elbow acts as a structural discontinuity, it could either act as a stress riser or as a structural fuse. In all cases examined, the peak elbow strains are less then the ground strain, except for pipe systems containing relatively thick walled pipe exposed to moderate wavelengths. Even in those cases, peak elbow strains only exceeded the ground strain by 25%. Furthermore, for the rather low levels of ground strain examined in the experiment by (Katayama), analytically predicted peak pipe strains (left side of Figures 3 –5) exceed those generated at the elbow, supporting the trends seen in the experiment itself.

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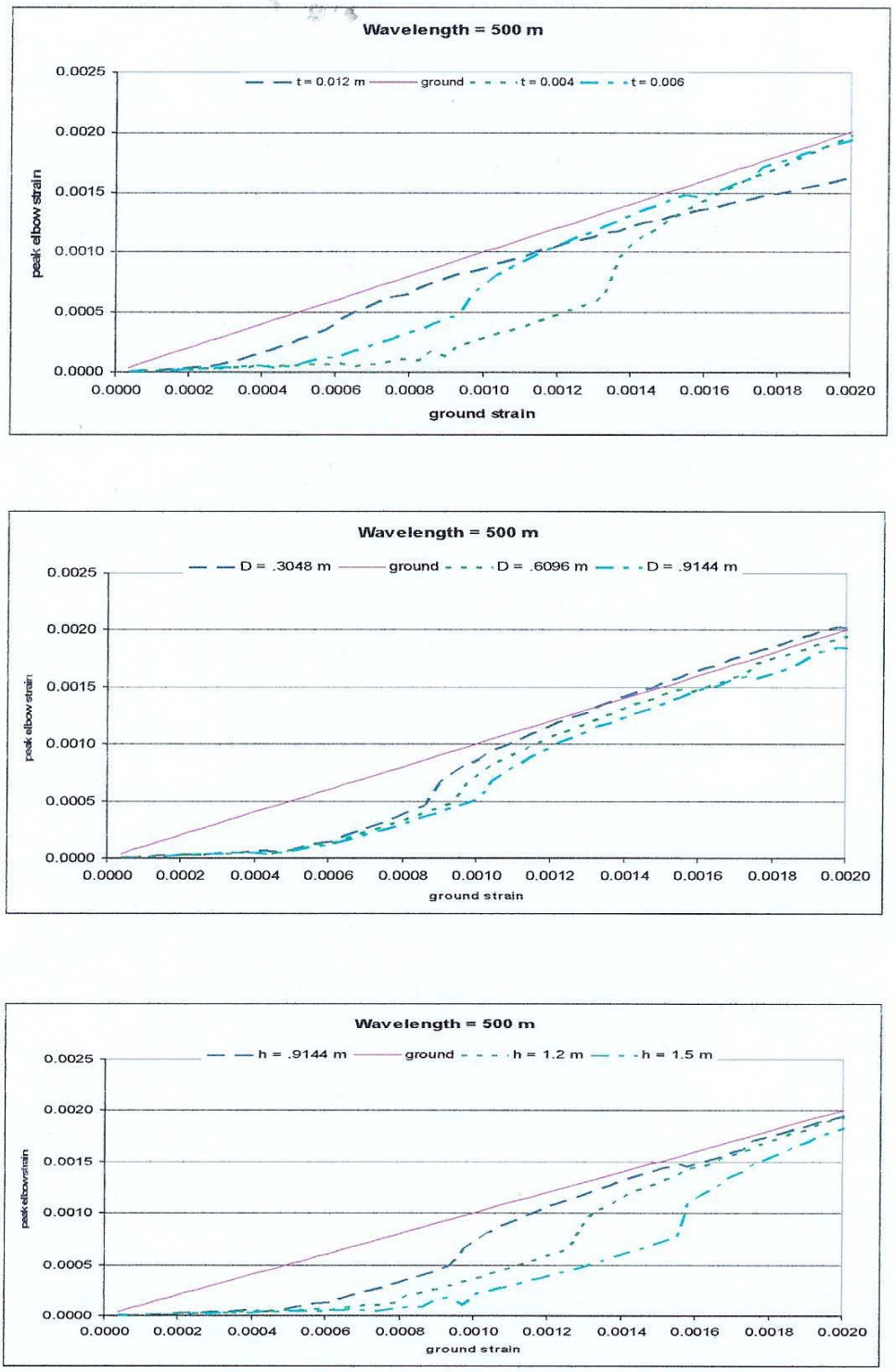

Figure 3. Variation of Pipe Elbow Strain with Respect to Ground Strain for Varying Pipe Wall Thickness, Pipe Diameter and Pipe Burial Depth for a Wavelength of 500m

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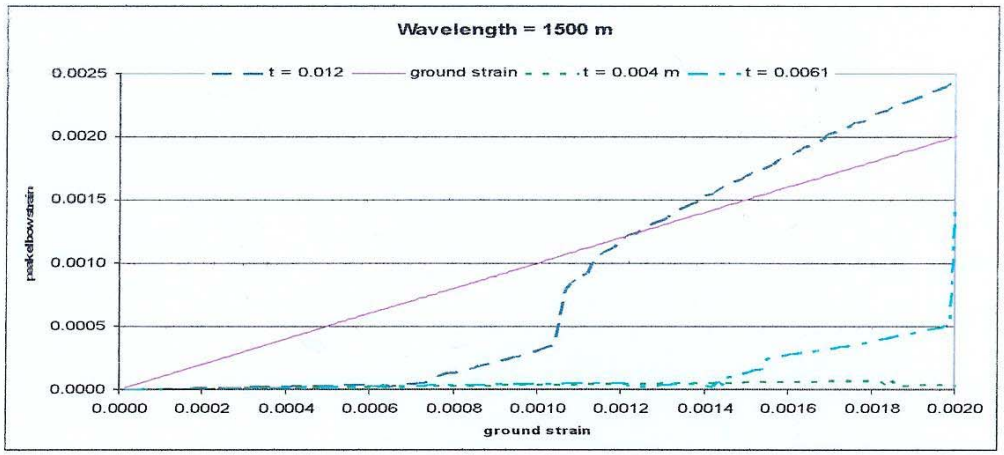

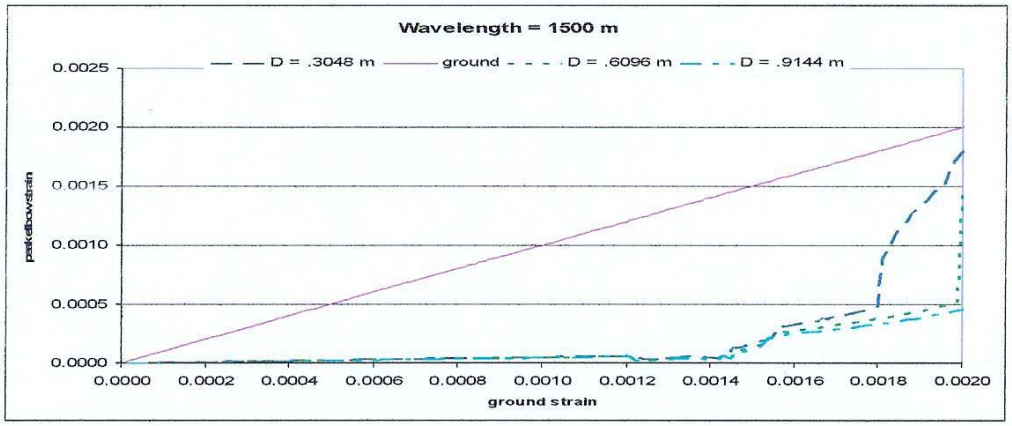

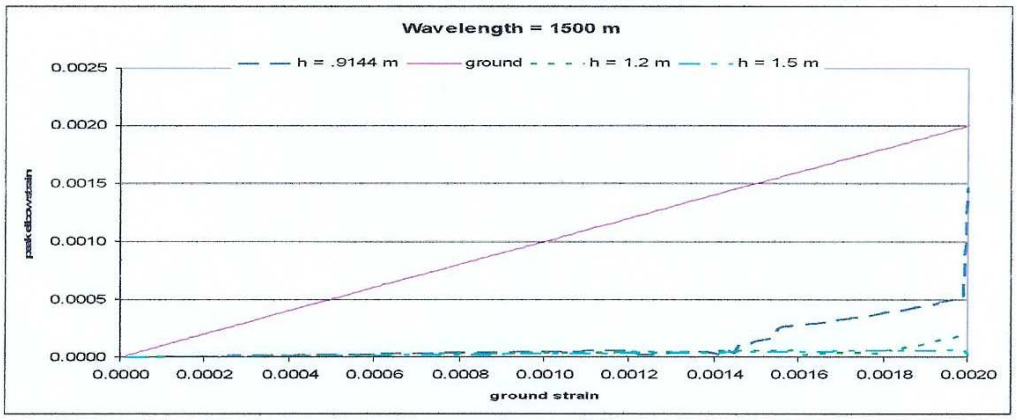

Figure 4. Variation of Pipe Elbow Strain with Respect to Ground Strain for Varying Pipe Wall Thickness, Pipe Diameter and Pipe Burial Depth for a Wavelength of 1500m

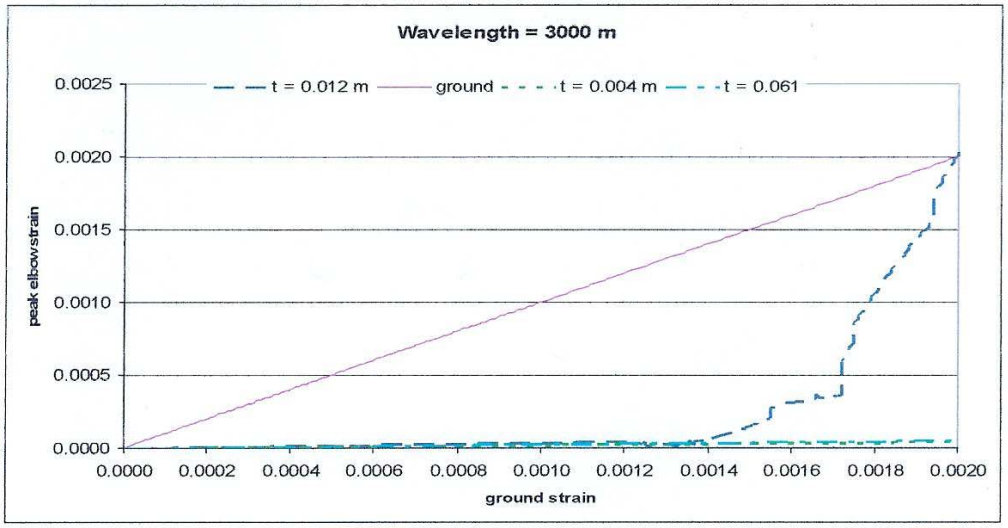

Figure 5. Variation of Pipe Elbow Strain with Respect to Ground Strain for Varying Pipe Wall Thickness for a Wavelength of 3000m

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